FULL PROJECT REPORT

Project: Asset Defender

Problem Statement

Social media users face persistent risks from fake profiles, impersonation campaigns, spam conversations, and phishing attempts. Most users cannot reliably identify sophisticated scam behavior, especially when attackers blend social engineering language with seemingly authentic profile metadata. Existing controls are either platform-side and opaque or lightweight local filters with low explainability. The project problem is to design and implement a practical system that detects profile and message risk in near real time, explains why a score was assigned, stores historical evidence for audit, and supports both end-user and administrator workflows without requiring enterprise-grade infrastructure.

Chapter 1 - Introduction

Asset Defender is a hybrid analytics platform for fake profile and scam message detection. The report captures architecture, implementation, and validation.

Chapter 2 - Objectives

- Detect fake profile behavior using structural and content features.
- Detect spam/scam messages using lexical and model-based signals.
- Provide explainable recommendations and confidence.
- Support user and admin dashboards with persistent history.

Chapter 3 - System Architecture

Frontend: React + Vite. Backend: Node.js + Express. Database: SQLite. ML: Python + scikit-learn models. Integration includes browser extension scripts.

Chapter 4 - Modules

1. Authentication module
2. Profile analysis module
3. Message analysis module
4. Behavior analysis module
5. Risk fusion module
6. Persistence module
7. Dashboard and admin analytics module

Chapter 5 - API Endpoints

POST /api/signup
POST /api/login
POST /api/admin/login
GET /api/admin/users
GET /api/admin/analyses
GET /api/admin/stats
GET /api/dashboard
POST /api/scrape
POST /api/analyses
POST /api/analyses/client
GET /api/analyses
GET /api/analyses/:id

Chapter 6 - Database Design

Tables: users, profile_analysis, message_analysis, behavior_analysis, final_predictions. Foreign key linkage by user_id.

Chapter 7 - ML and Heuristics

Profile models: RF, GB, LR, XGBoost. Message models: MultinomialNB and LR with TF-IDF. Heuristic analyzers compute risk features from bio, username, captions, comments, and message patterns.

Chapter 8 - Evaluation

Model Evaluation Snapshot

- Domain: profile, Model: random_forest, Accuracy: 0.9310, Precision: 0.9167, Recall: 0.9483, F1: 0.9322, ROC-AUC: 0.9854
- Domain: profile, Model: gradient_boosting, Accuracy: 0.9224, Precision: 0.9153, Recall: 0.9310, F1: 0.9231, ROC-AUC: 0.9834
- Domain: profile, Model: logistic_regression, Accuracy: 0.9483, Precision: 0.9815, Recall: 0.9138, F1: 0.9464, ROC-AUC: 0.9887
- Domain: profile, Model: xgboost, Accuracy: 0.9310, Precision: 0.9310, Recall: 0.9310, F1: 0.9310, ROC-AUC: 0.9831
- Domain: message, Model: multinaive, Accuracy: 0.9740, Precision: 0.9918, Recall: 0.8121, F1: 0.8930, ROC-AUC: 0.9904
- Domain: message, Model: message_logistic_regression, Accuracy: 0.9614, Precision: 1.0000, Recall: 0.7114, F1: 0.8314, ROC-AUC: 0.9871

Chapter 9 - Security

bcrypt password hashing, JWT token authentication, admin role gate, CORS allowlist, security headers, rate limits, and login throttling.

Chapter 10 - Conclusion

The project delivers a practical and extensible anti-scam analysis platform, with clear opportunities for multilingual support, model registry, drift monitoring, and stronger CI/CD automation.